Configuration

The KUCCPS IT Ticketing System uses environment variables for configuration. Never commit real credentials to version control.

Backend Configuration

Create a `.env` file in the `backend/` directory.

Database

```
DATABASE_URL="postgresql://user:password@localhost:5432/database_name?schema=public"
```

Server

```
PORT=5000
NODE_ENV=development
```

JWT Authentication

```
JWT_SECRET=change-this-to-a-long-random-secret
JWT_EXPIRES_IN=7d
```

- `JWT_SECRET` must be at least 32 characters in production.

CORS

```
FRONTEND_URL=http://localhost:5173,http://localhost:5174
```

For local development, both Vite ports are allowed. In production, use HTTPS URLs.

Public Ticket API Key

```
PUBLIC_TICKET_API_KEY=change-this-to-a-long-random-secret
REQUIRE_PUBLIC_TICKET_API_KEY=true
```

- Used by the Google Workspace Add-on to create tickets.
- Must be at least 32 characters in production.

SMTP / Outbound Email

```
EMAIL_SERVICE=gmail
EMAIL_USER=
EMAIL_PASSWORD=
EMAIL_FROM="KUCCPS IT Support <itsupport@example.ac.ke>"
SUPPORT_EMAIL=itsupport@example.ac.ke
```

IMAP / Inbound Email Monitoring

```
IMAP_HOST=imap.gmail.com
IMAP_PORT=993
IMAP_USER=
IMAP_PASSWORD=
IMAP_TLS=true
```

Email monitoring is disabled unless `EMAIL_USER` and `IMAP_USER` are configured.

LDAP / Active Directory

```
USE_LDAP_AUTH=false
LDAP_URL=ldaps://ldap.example.ac.ke:636
LDAP_BIND_DN=
LDAP_BIND_PASSWORD=
LDAP_SEARCH_BASE=
LDAP_SEARCH_FILTER=(uid={{username}})
```

- Use `ldaps://` (LDAPS) in production, never plain `ldap://`.

File Uploads

```
MAX_FILE_SIZE=10485760
MAX_UPLOAD_BYTES=10485760
MAX_KB_UPLOAD_BYTES=5242880
UPLOAD_DIR=./uploads
```

Production Hardening

```
ALLOW_PUBLIC_API_KEY_IN_BODY=false
SMTP_TLS_REJECT_UNAUTHORIZED=true
IMAP_TLS_REJECT_UNAUTHORIZED=true
LDAP_TLS_REJECT_UNAUTHORIZED=true
JSON_BODY_LIMIT=1mb
```

Frontend Configuration

Create a `.env` file in the `frontend/` directory.

```
VITE_API_URL=http://localhost:5000/api
```

For production:

```
VITE_API_URL=https://api.example.ac.ke/api
```

Environment Variable Reference

Variable	Purpose	Required	Default
`DATABASE_URL`	PostgreSQL connection string	Yes	—
`PORT`	Backend server port	No	`5000`
`NODE_ENV`	Runtime environment	No	`development`
`JWT_SECRET`	JWT signing secret	Yes	—
`JWT_EXPIRES_IN`	Token expiration	No	`7d`
`FRONTEND_URL`	Allowed CORS origins	No	`http://localhost:5173,http://localhost:5174`
`PUBLIC_TICKET_API_KEY`	Google Add-on API key	Yes	—

Variable	Purpose	Required	Default
`REQUIRE_PUBLIC_TICKET_API_KEY`	Enforce API key check	No	`true`
`EMAIL_SERVICE`	SMTP provider	No	`gmail`
`EMAIL_USER`	SMTP username	Conditional	—
`EMAIL_PASSWORD`	SMTP password	Conditional	—
`EMAIL_FROM`	Sender display name	No	`KUCCPS IT Support`
`SUPPORT_EMAIL`	Support mailbox	No	`itsupport@kuccps.ac.ke`
`IMAP_HOST`	IMAP server host	Conditional	`imap.gmail.com`
`IMAP_PORT`	IMAP server port	Conditional	`993`
`IMAP_USER`	IMAP username	Conditional	—
`IMAP_PASSWORD`	IMAP password	Conditional	—

Variable	Purpose	Required	Default
`USE_LDAP_AUTH`	Enable LDAP login	No	`false`
`LDAP_URL`	LDAP server URL	Conditional	—
`MAX_FILE_SIZE`	Max upload size in bytes	No	`10485760`
`UPLOAD_DIR`	Upload storage path	No	`./uploads`
`VITE_API_URL`	Frontend API base URL	Yes	—

Configuration Best Practices

1. **Never commit `.env` files** — they are gitignored by default.
2. **Rotate secrets before production** — generate new `JWT_SECRET` and `PUBLIC_TICKET_API_KEY`.
3. **Use HTTPS in production** — update `FRONTEND_URL` and `VITE_API_URL` accordingly.
4. **Use LDAPS, not LDAP** — plain LDAP sends credentials in cleartext.
5. **Configure production CORS explicitly** — avoid wildcard origins.

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